

CONCEPT MAP · GENERATIVE AI

Architecture & Information Flow

Name: Murthi Haari **Reg No:** 192472050**Course:** CSA6503 – Generative AI and Large Scale Models**Transformer Encoder**

A stack of identical layers that converts input tokens into rich contextual representations.

Receives

**1. Input Embedding****Definition:** Converts each input token (word/sub-word) into a dense numerical vector.**Purpose:** Represents word meaning in a continuous vector space the model can process.**Note:** Embedding dimension is typically 512 or 768.

Adds

**2. Positional Encoding****Definition:** Injects information about token order using sine/cosine functions.**Purpose:** Provides sequence order to the model, since self-attention has no built-in notion of position.**Note:** Added (not concatenated) to the input embedding.

Provides sequence order to

**3. Multi-Head Self-Attention**

Definition: Computes attention scores between every pair of tokens using multiple parallel attention heads.

Purpose: Captures relationships and dependencies between tokens regardless of distance.

Note: Multiple heads let the model attend to different aspects (syntax, meaning) simultaneously.

Captures relationships between

4. Add & Layer Normalization

Definition: Adds the attention output to the original input (residual connection) and normalizes it.

Purpose: Stabilizes training and prevents vanishing/exploding gradients.

Note: Also called a "residual + norm" block; used twice per encoder layer.

Stabilizes and passes to

5. Feed Forward Neural Network

Definition: A position-wise fully connected network (two linear layers with a ReLU/GELU activation) applied to each token independently.

Purpose: Learns complex, non-linear feature representations beyond what attention alone provides.

Note: The hidden layer is usually 4× wider than the embedding dimension.

Refines features via

6. Add & Layer Normalization

Definition: Adds the feed-forward output back to its input (residual connection) and normalizes it again.

Purpose: Further stabilizes training and preserves information flow across layers.

Note: Completes one full encoder layer/block.

Produces

7. Encoder Output

Definition: The final set of context-rich vectors representing every input token.

Purpose: Serves as input to the decoder (or downstream task head) for further processing.

Note: Same shape as the input embeddings, but now context-aware.

 **Stacking Note:** Steps 3–6 form **one Encoder Block**. Real Transformer models stack **N identical blocks** (e.g., N = 6 in the original paper, N = 12/24 in BERT) one after another for deeper representation learning.

💡 The Transformer Encoder processes all input tokens simultaneously using self-attention — enabling parallelization and long-range context capture.

REAL-WORLD APPLICATIONS

BERT

ChatGPT (Encoder concepts)

Machine Translation

Text Classification

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